Learning Power BI the Hands-On Way: My Data Visualization Lab Experience

For my Data Visualization Lab, I had to work on the Adventure Works dataset in Power BI. If you have never used Power BI before, don't worry, I am going to explain everything in simple words. No jargon, just plain explanations like I would tell a friend.

First, what even is Power BI?

Power BI is a tool made by Microsoft that helps you take raw data (think of a big Excel sheet with thousands of rows) and turn it into something useful, like charts, numbers, and reports. But before you can make nice charts, you usually need to clean up and organize your data. That part is done inside something called the Power Query Editor.

Think of Power Query as the kitchen where you prep your ingredients before cooking. You wash the vegetables, cut them properly, remove the bad parts. Only after that do you actually cook the meal (which in Power BI terms means building your dashboard).

My practical had three parts, so let me walk you through what I did in each one.

Part A: Just Adding Up Numbers (Aggregations)

The first part was about aggregations. That's just a fancy word for basic math operations on a column of data, like adding numbers, finding the average, or counting rows.

I worked with functions like:

- SUM, which just adds everything up. I used it to find the total product price and the total quantity sold.
- AVERAGE, which finds the mean value. I used it on product price to see what a "typical" price looks like.
- COUNT, which counts how many rows there are. I used this to count the total orders.
- DISTINCT COUNT, which counts only unique values. This one matters a lot, because if I just used COUNT on customers, I might count the same customer multiple times if they ordered more than once. DISTINCT COUNT fixes that and gives the actual number of different customers.
- MIN and MAX, which find the smallest and largest values. I used these on product cost to see the cheapest and the most expensive product.

This part was honestly the easiest. It's basically the same math we learned in school, just applied to a big dataset instead of a small list of numbers.

Part B: Creating New Columns (Calculated Columns)

This part had thirty small tasks and took the most time. The idea here is that sometimes your original data doesn't have exactly the column you need, so you create a new one using Power Query.

I'll group what I did into a few categories so it's easier to follow.

Getting date information out of a date column A date like 15/01/2024 actually hides a lot of information. I extracted the year, the month name, the quarter, the week number, and even the day name from it. Power Query has ready-made options for all of this under Add Column, so you don't have to calculate anything manually.

Cleaning up messy text Real data is rarely perfectly typed. Some product names might have extra spaces, wrong capital letters, or weird characters. I used tools to trim spaces, convert text to uppercase or lowercase, and clean up any junk characters. I also capitalized the first letter of every word to make names look neat, like "Mountain Bike" instead of "mountain bike".

Splitting and merging columns Sometimes one column actually has two pieces of information stuck together, so I split it. For example, splitting a full name into first name and last name. Other times I did the opposite, merging two columns into one, like combining product category and subcategory into a single label.

Making decisions with conditional columns This was my favorite part. A conditional column lets you say "if this condition is true, put this value, otherwise put that value." I used this to create a Price Range column that labels things as Low, Medium or High depending on the price, and a Profit Category column that flags whether a product made a loss or a profit. It's basically an if-else statement, but done without writing any code.

Grouping data together I also used the Group By feature, which is similar to a pivot table. For example, I grouped all products by category and added up the total sales for each category, so instead of seeing every single row, I get one summary row per category.

Small fixes The rest were smaller cleanup tasks like removing duplicate customer rows, sorting products by sales, rounding off decimal values, and adding an index column, which just numbers each row from 1 onwards like a serial number.

Part C: Real Business-Style Tasks

The last part combined everything I learned above into more practical, real world style tasks. This is basically how someone working with sales data at a company would actually use Power BI.

I built a Sales Band column that groups every order into Low, Medium or High based on how much it was worth. I made a Shipping Performance column that checks how many days an order took to ship and labels it as Fast, Normal or Delayed. I also created a Discount Band,

extracted a combined Month-Year field like "Jan-2024", and grouped all the data by region to see total sales, average sales, and total quantity for each region.

This part felt the most rewarding because I could actually see how these small techniques come together to answer real questions, like which region is performing well or how fast orders are being shipped.

What I actually learned

Before this practical, I honestly thought Power BI was just about making pretty charts. Turns out, most of the real work happens before the chart is even built, in cleaning, organizing and shaping the data properly. Power Query does a lot of heavy lifting, and once your data is in good shape, actually visualizing it becomes the easy part.

If you're a beginner trying to learn Power BI, my honest suggestion is to not skip the Power Query part. It feels a bit boring compared to building colorful charts, but it's where you actually learn how to think about data.